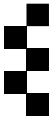
Data Warehouses



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Data Warehouses (1)

- Data warehouses are database systems that have
  - data
  - Programs
  - and personnel *(specialize in the preparation of data)*

  for BI processing.
- Data warehouse databases differ from operational databases because the **data warehouse data is frequently denormalized**.

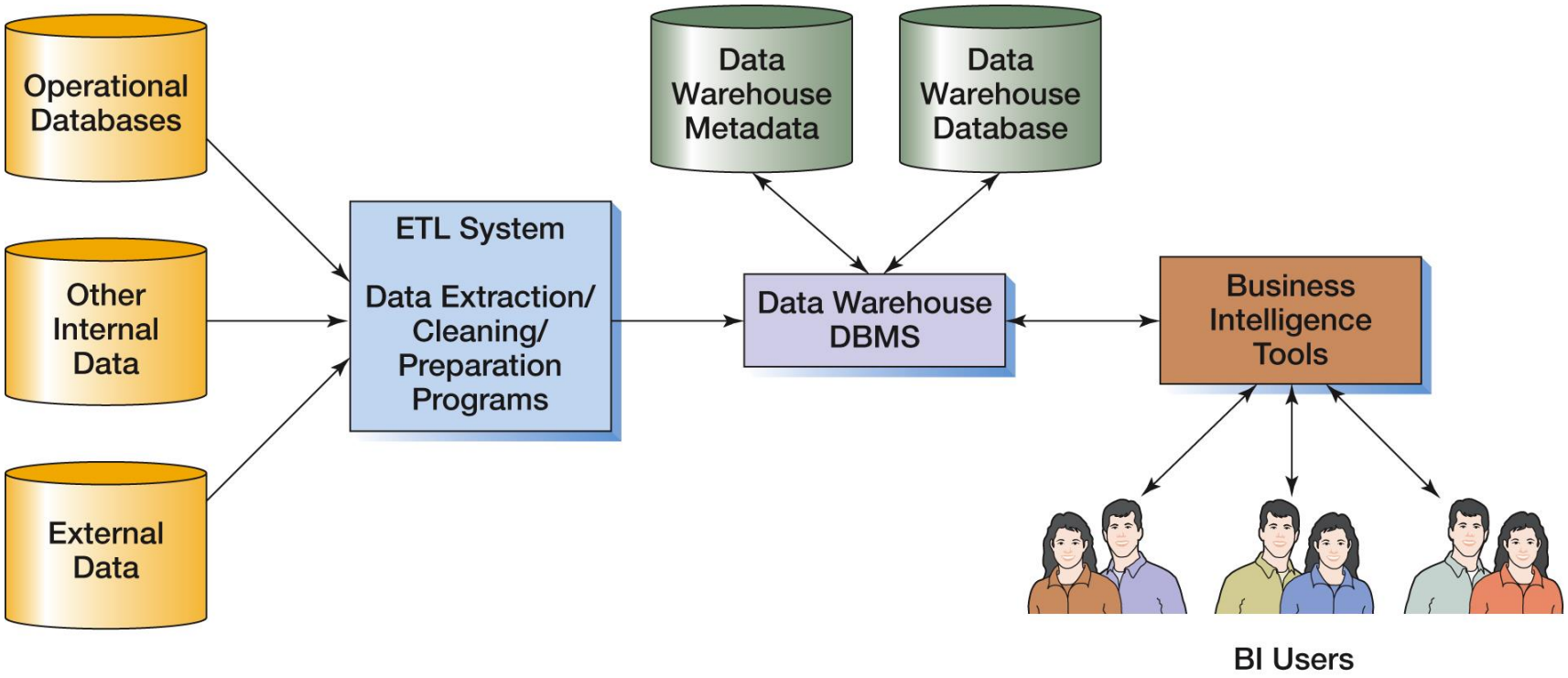
Data Warehouses (2)

- Further, that data is **never inserted, updated, or deleted by users**, but only by data warehouse administrators.
- Data warehouses vary in **scale and scope**.
- They can be as simple as **a sole employee** processing a data extract on a part-time basis

  OR

- as **complex as a department** with dozens of employees maintaining libraries of data and programs.

Components of a Data Warehouse



Components of a Data Warehouse
ETL System (1)

- **Data are read** from operational databases by the Extract, Transform, and Load (**ETL**) system.
- The ETL system then **cleans and prepares** the data for BI processing.
- This can be a **complex process**.
  - First, the data may be **problematic**.

Components of a Data Warehouse
ETL System (2)

- Second, data may need to be **changed or transformed** for use in a data warehouse.

  For example, the operational systems may store data about countries using standard two-letter country codes, such as US (United States) and CA(Canada).

- However, applications using the data warehouse may **need to use the country names in full**.
- Thus, the data transformation ***CountryCode* to *CountryName*** will be needed before the data can be loaded into the data warehouse

Components of a Data Warehouse
ETL System (3)

- The ETL stores the extracted data in a data warehouse database using a **data warehouse DBMS**, which can be **different** from the organization's **operational DBMS**.
- **For example,** an organization might use Oracle Database for its operational processing but use Microsoft SQL Server 2014 for its data warehouse.

Components of a Data Warehouse Metadata

- Metadata concerning
  - data's source
  - Format
  - assumptions and constraints,
  - and other facts

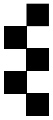
are kept in a **data warehouse metadata database**.

- The data warehouse DBMS extracts and provides data to BI tools, such as data mining programs.

Briefly

- You can think of a data warehouse as a **distributor in a supply chain.**
- The data warehouse takes data from the data manufacturers (**operational systems and purchased data**), cleans and processes them, and locates the data on the shelves of the data warehouse.
- The people who work in a data warehouse are experts at data management, data cleaning, and data transformation.
- However, they are not usually experts in a given business function.

Data Mart and Enterprise Data Warehouse



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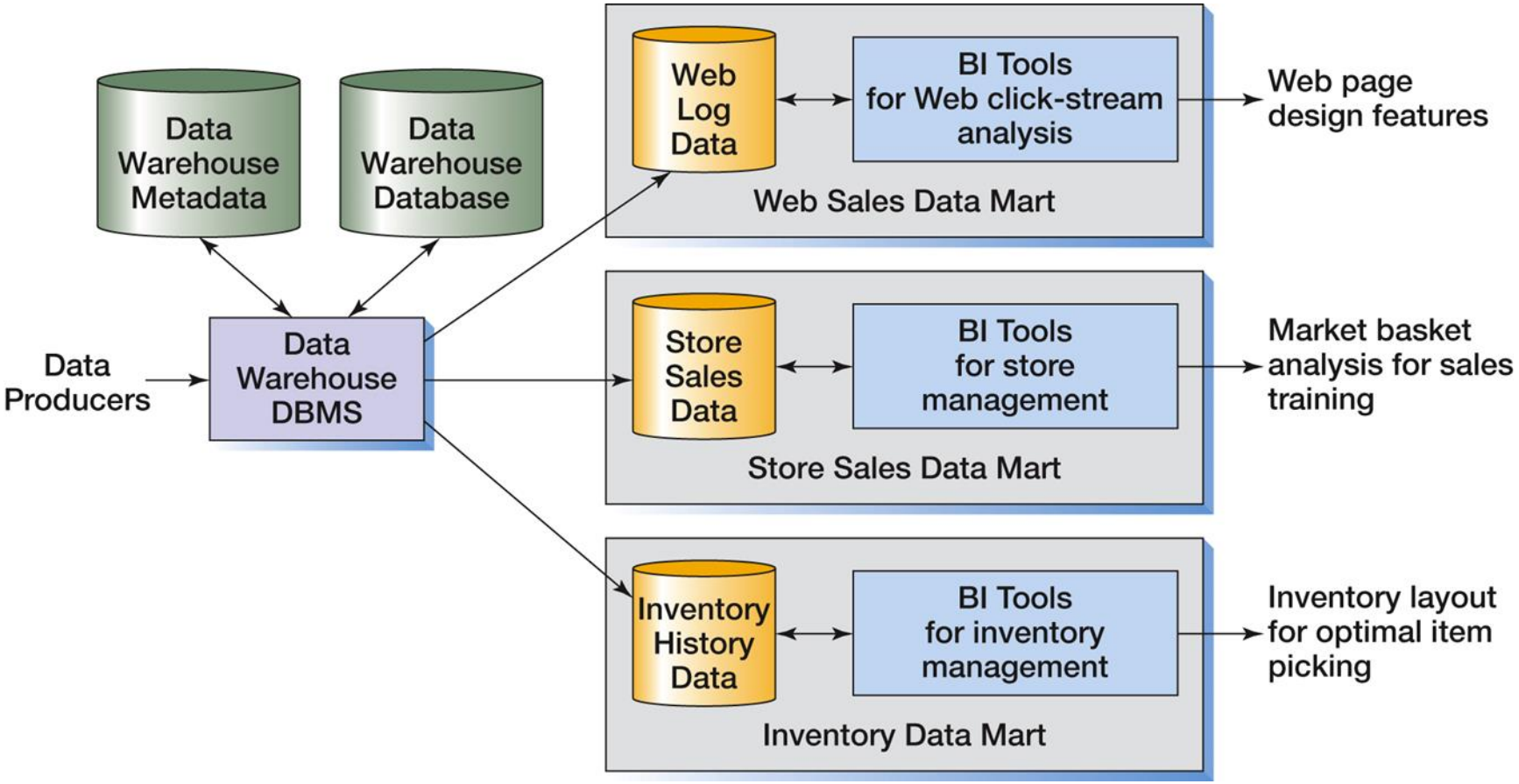
Data Mart

- A data mart is a collection of data that is **smaller** than that in the data warehouse

and

- that addresses a **particular component** or functional area of the business.
- A data mart is like a retail store in a supply chain.

Data Warehouse and Data Marts (1)



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Data Warehouse and Data Marts (2)

This data warehouse takes data from the data producers and distributes the data to three data marts.

- One data mart **analyzes click-stream** data
  - for the purpose of designing Web pages.
- The second **analyzes store sales data** and determines which products tend to be purchased together.
  - This information is used to train salespeople on the best way to up-sell customers.

Data Warehouses and Data Marts (3)

- The third data mart analyzes **customer order data** for the purpose of reducing labor for item picking from the warehouse.
  - A company such as Amazon.com, for example, goes to great lengths to organize its warehouses to reduce picking expenses.

Enterprise Data Warehouse

- When the **data mart structure is combined with the data warehouse architecture**, the combined system is known as an enterprise data warehouse (EDW) architecture.
- In this configuration, the data warehouse maintains all enterprise BI data and acts as the **authoritative source** for data extracts provided to the data marts.
- The data marts receive all their data from the data warehouse—they **do not add or maintain any additional data**.

Transient versus Periodic Data

- **Transient** – changes to existing records are written over previous records, destroying previous data content

- **Periodic** – data is never physically altered or deleted after being added to the store

Transient Operational Data

Table X (10/09)

Key	A	B
001	a	b
002	c	d
003	e	f
004	g	h

Table X (10/10)

Key	A	B
001	a	b
002	r	d
003	e	f
004	y	h
005	m	n

Table X (10/11)

Key	A	B
001	a	b
002	r	d
003	e	t
005	m	n

sex

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Periodic Warehouse Data

Table X (10/09)

Key	Date	A	B	Action
001	10/09	a	b	C
002	10/09	c	d	C
003	10/09	e	f	C
004	10/09	g	h	C

Table X (10/10)

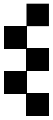
Key	Date	A	B	Action
001	10/09	a	b	C
002	10/09	c	d	C
002	10/10	r	d	U
003	10/09	e	f	C
004	10/09	g	h	C
004	10/10	y	h	U
005	10/10	m	n	C

Table X (10/11)

Key	Date	A	B	Action
001	10/09	a	b	C
002	10/09	c	d	C
002	10/10	r	d	U
003	10/09	e	f	C
003	10/11	e	t	U
004	10/09	g	h	C
004	10/10	y	h	U
004	10/11	y	h	D
005	10/10	m	n	C

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Dimensional Databases



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Dimensional Database

- The databases in a data warehouse or data mart are built to a **different type of database design**.
- The data warehouse databases are built in a design called **a dimensional database** that is designed for efficient data queries and analysis.
- A dimensional database is used to **store historical data** rather than just the current data stored in an operational database.

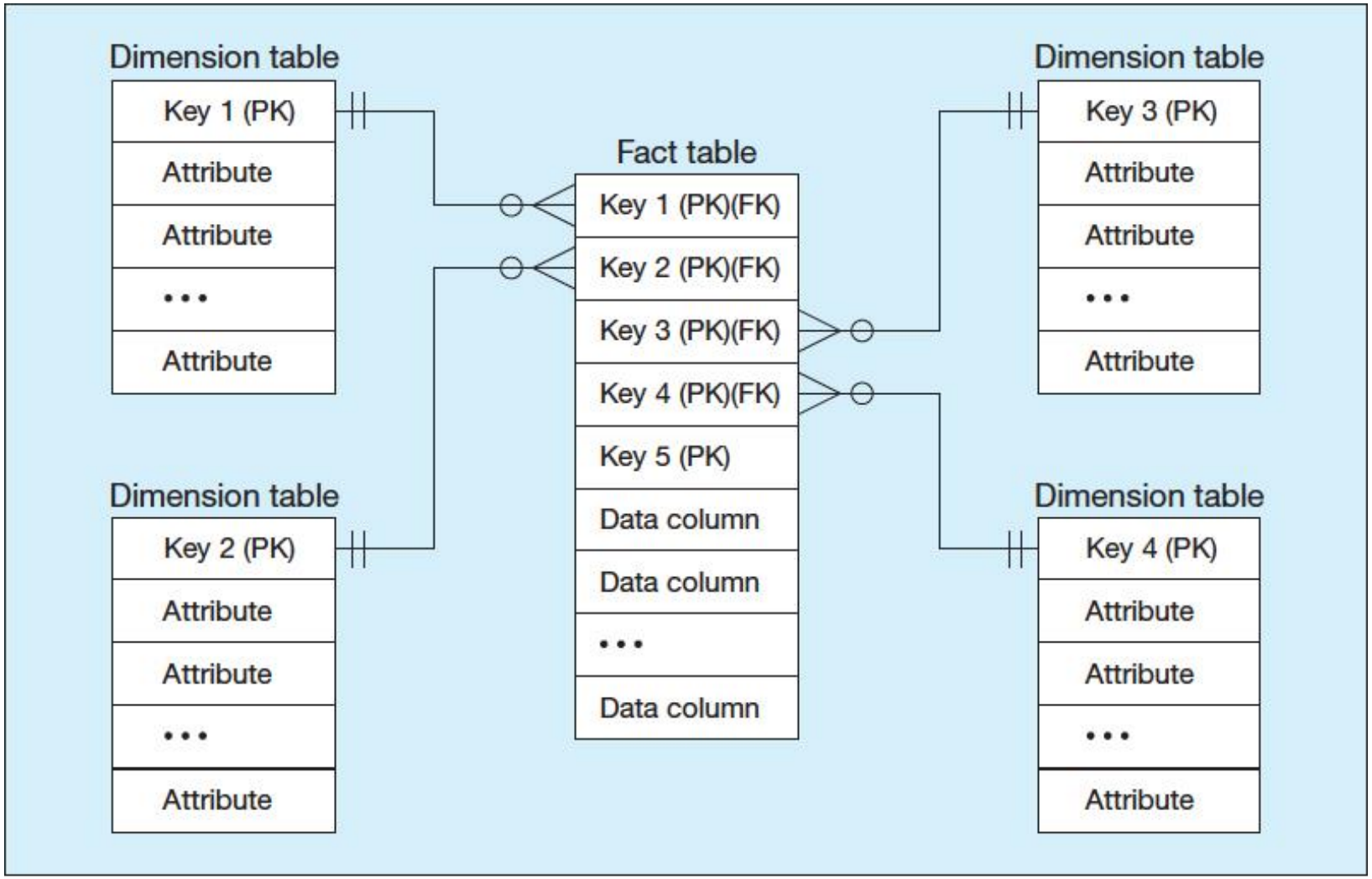
Star Schema (1)

- A star schema, so named because it visually **resembles a star**, has a fact table at the center of the star and dimension tables radiating out from the center.
- The fact table is always fully normalized, but dimension tables may be non-normalized.
- Fact tables contain time-period data, so **date dimensions** are important.
- Dimension table keys should be **surrogate.**

Star Schema (2)

- Each dimension table has a **one-to-many relationship** to the central fact table.
- Each dimension table generally has a **simple primary key** as well as several non-key attributes. The primary key, in turn, is a **foreign key in the fact table**.
- The **primary key of the fact table is a composite key** that consists of the concatenation of all of the foreign keys (plus possibly other components that do not correspond to dimensions.

Star Schema (3)



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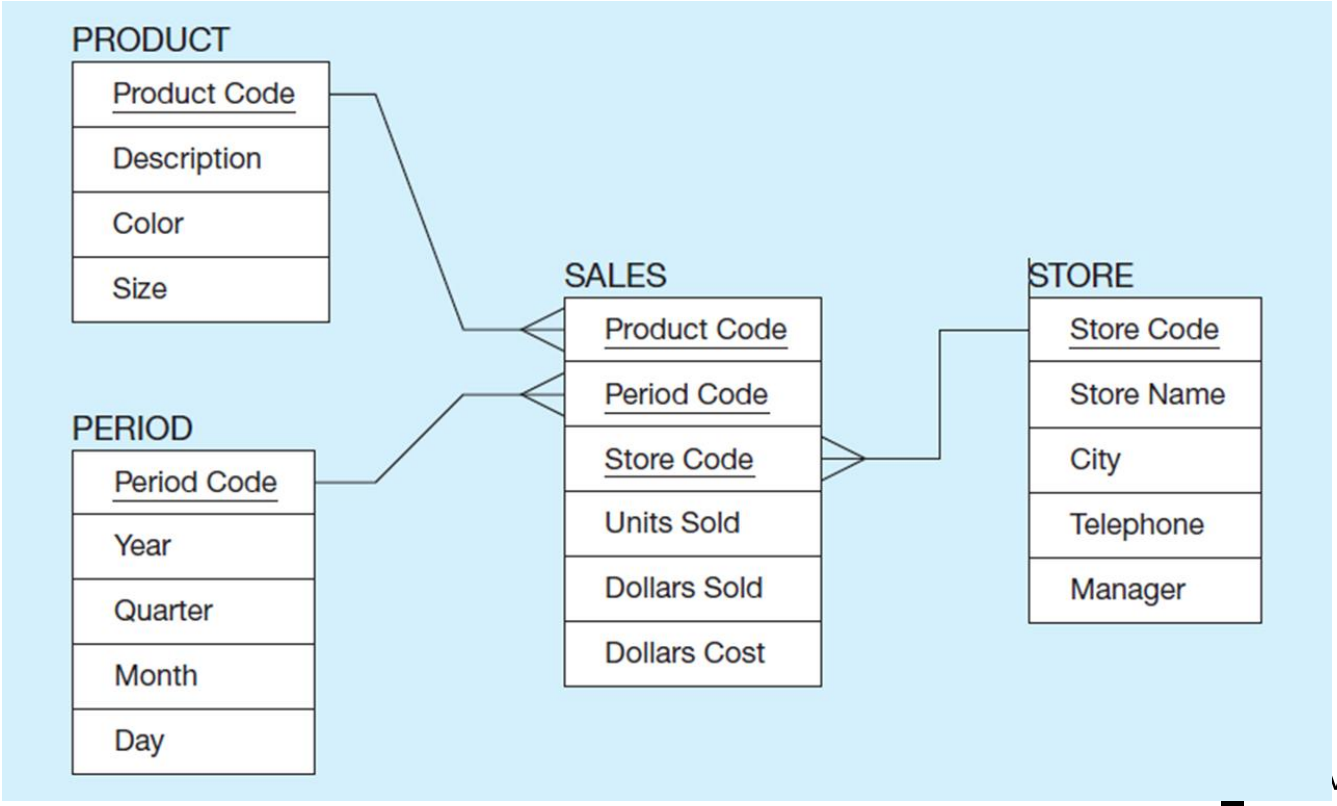
Fact Tables and Dimension Tables

Fact tables contain **factual or quantitative data** (measurements that are numerical, continuously valued, and additive) about a business, such as units sold, orders booked, and so forth.

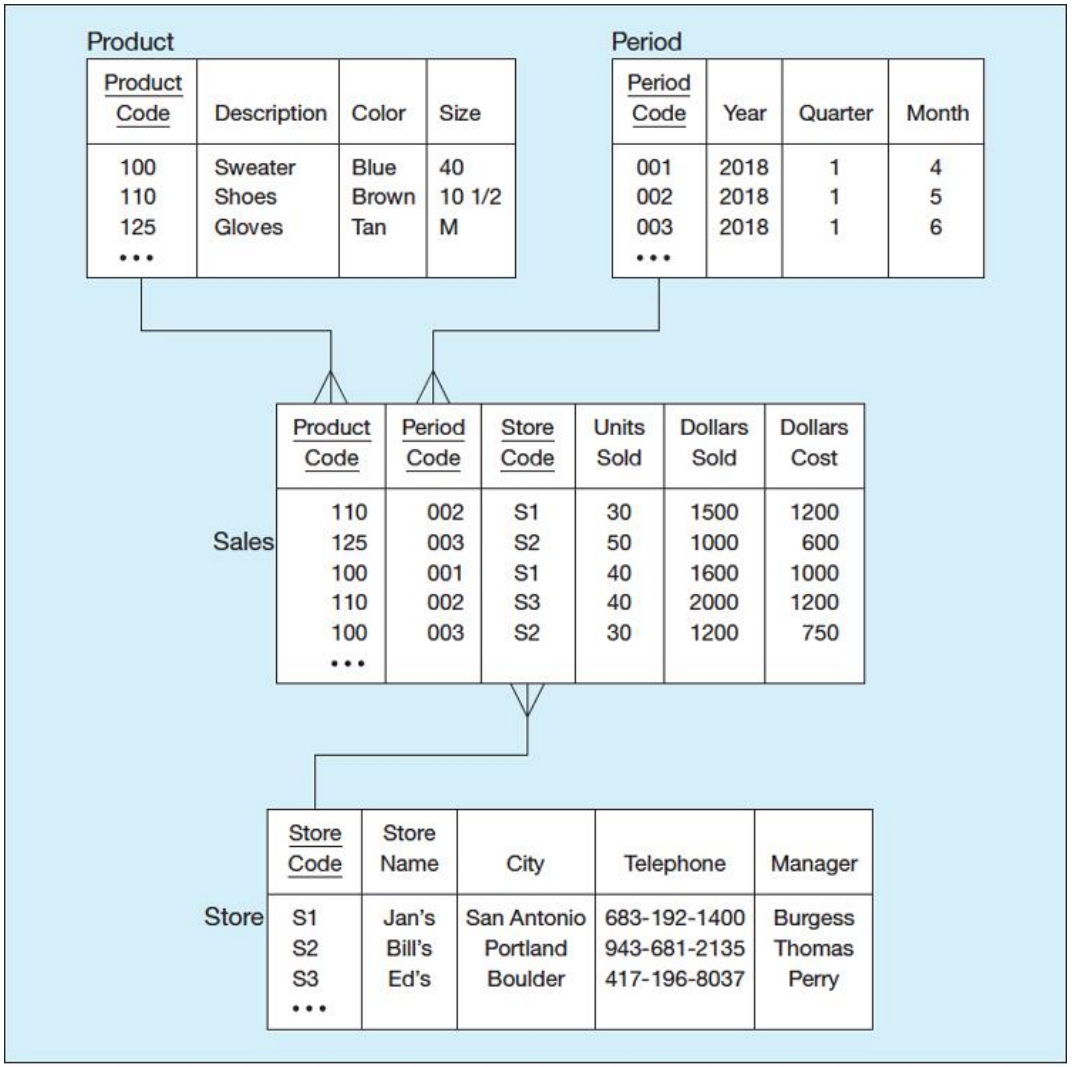
Dimension tables hold **descriptive data** (context) about the subjects of the business. The dimension tables are usually the **source of attributes** used to qualify, categorize, or summarize facts in queries, reports, or graphs; thus, dimension data are usually textual and discrete (even if numeric).

Example of a Star Schema

Fact table **provides statistics** for sales broken down by product, period, and store dimensions



Star Schema with Sample Data



Product

Product Code	Description	Color	Size
100	Sweater	Blue	40
110	Shoes	Brown	10 1/2
125	Gloves	Tan	M
...			

Period

Period Code	Year	Quarter	Month
001	2018	1	4
002	2018	1	5
003	2018	1	6
...			

Sales

Product Code	Period Code	Store Code	Units Sold	Dollars Sold	Dollars Cost
110	002	S1	30	1500	1200
125	003	S2	50	1000	600
100	001	S1	40	1600	1000
110	002	S3	40	2000	1200
100	003	S2	30	1200	750
...					

Store

Store Code	Store Name	City	Telephone	Manager
S1	Jan's	San Antonio	683-192-1400	Burgess
S2	Bill's	Portland	943-681-2135	Thomas
S3	Ed's	Boulder	417-196-8037	Perry
...				

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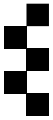
Use of Star Schema

A star schema provides answers to a domain of business questions.

For example, consider the following questions:

1. Which cities have the highest sales of large products?
2. What is the average monthly sales for each store manager?
3. On which stores and which products is the company losing money? Does this vary by quarter?

How to Create a Star Schema from a Relational Schema



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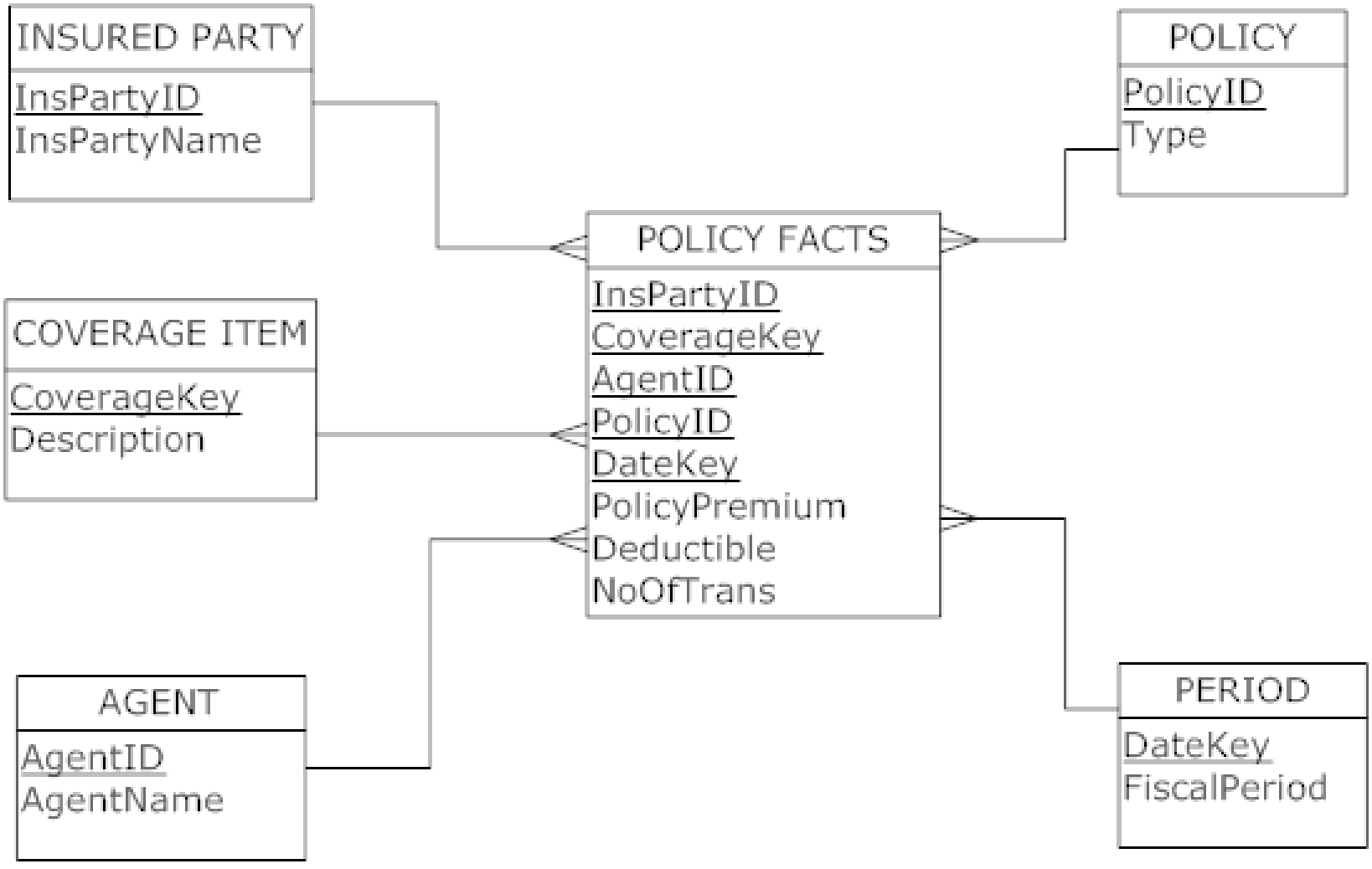
Exercise 1: Construct a Star Schema for Simplified Auto-mobile Insurance Company

The relevant dimensions, dimension attributes, and dimension sizes are as follows

InsuredParty	**Attributes:** InsuredPartyID and Name. There is an average of two insured parties for each policy and covered item
CoverageItem	**Attributes:** CoverageKey and Description. There is an average of 10 covered items per policy.
Agent	**Attributes:** AgentID and AgentName. There is one agent for each policy and covered item.
Policy	**Attributes:** PolicyID and Type. The company has approximately 1 million policies at the present time.
Period	**Attributes:** DateKey and FiscalPeriod

Facts to be recorded for each combination of these dimensions are PolicyPremium, Deductible, and NumberOfTransactions.

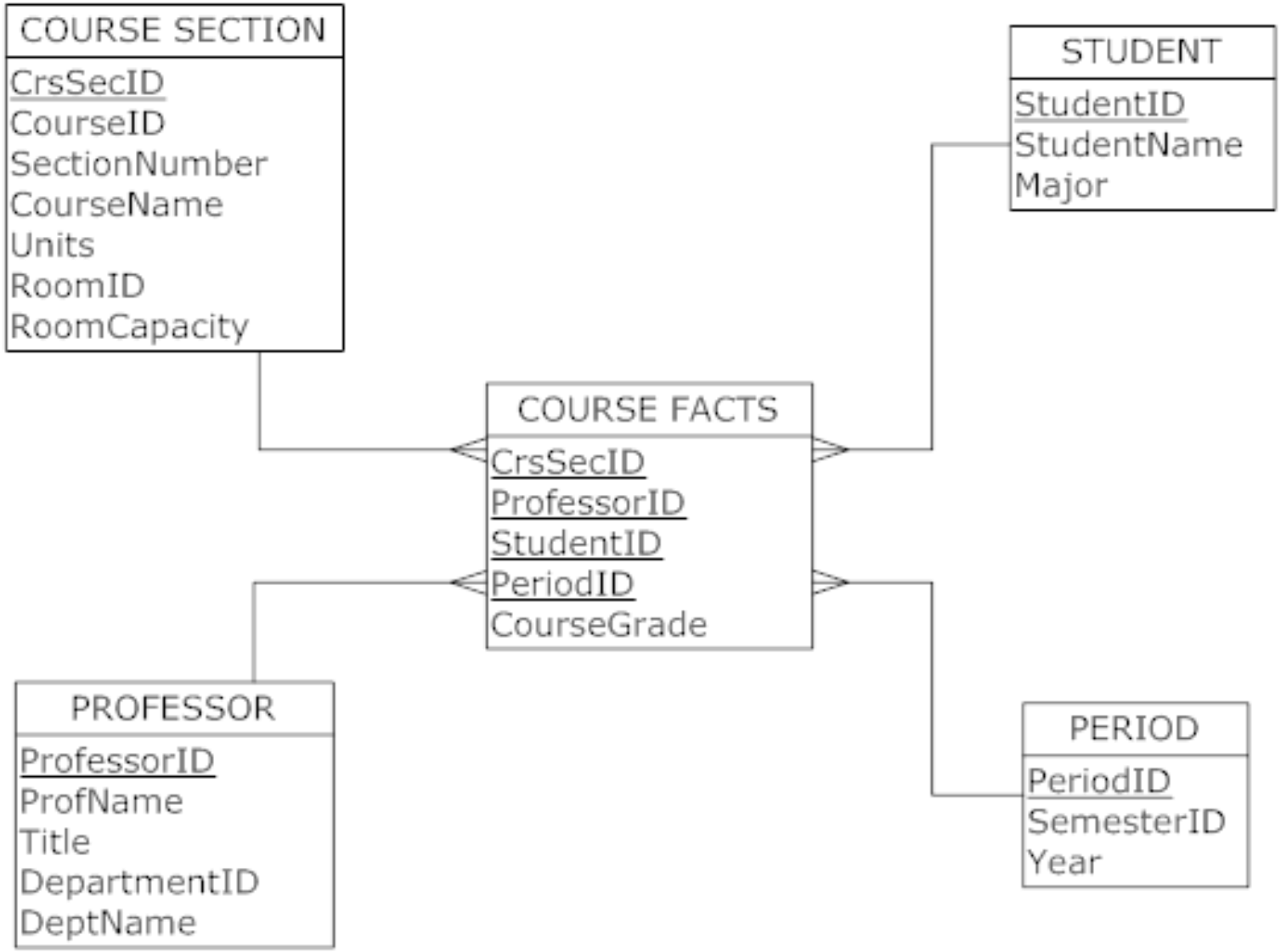
Solution


INSURED PARTY
InsPartyID
InsPartyName
POLICY
PolicyID
Type
POLICY FACTS
InsPartyID
CoverageKey
AgentID
PolicyID
DateKey
PolicyPremium
Deductible
NoOfTrans
COVERAGE ITEM
CoverageKey
Description
AGENT
AgentID
AgentName
PERIOD
DateKey
FiscalPeriod

Snowflake Schema

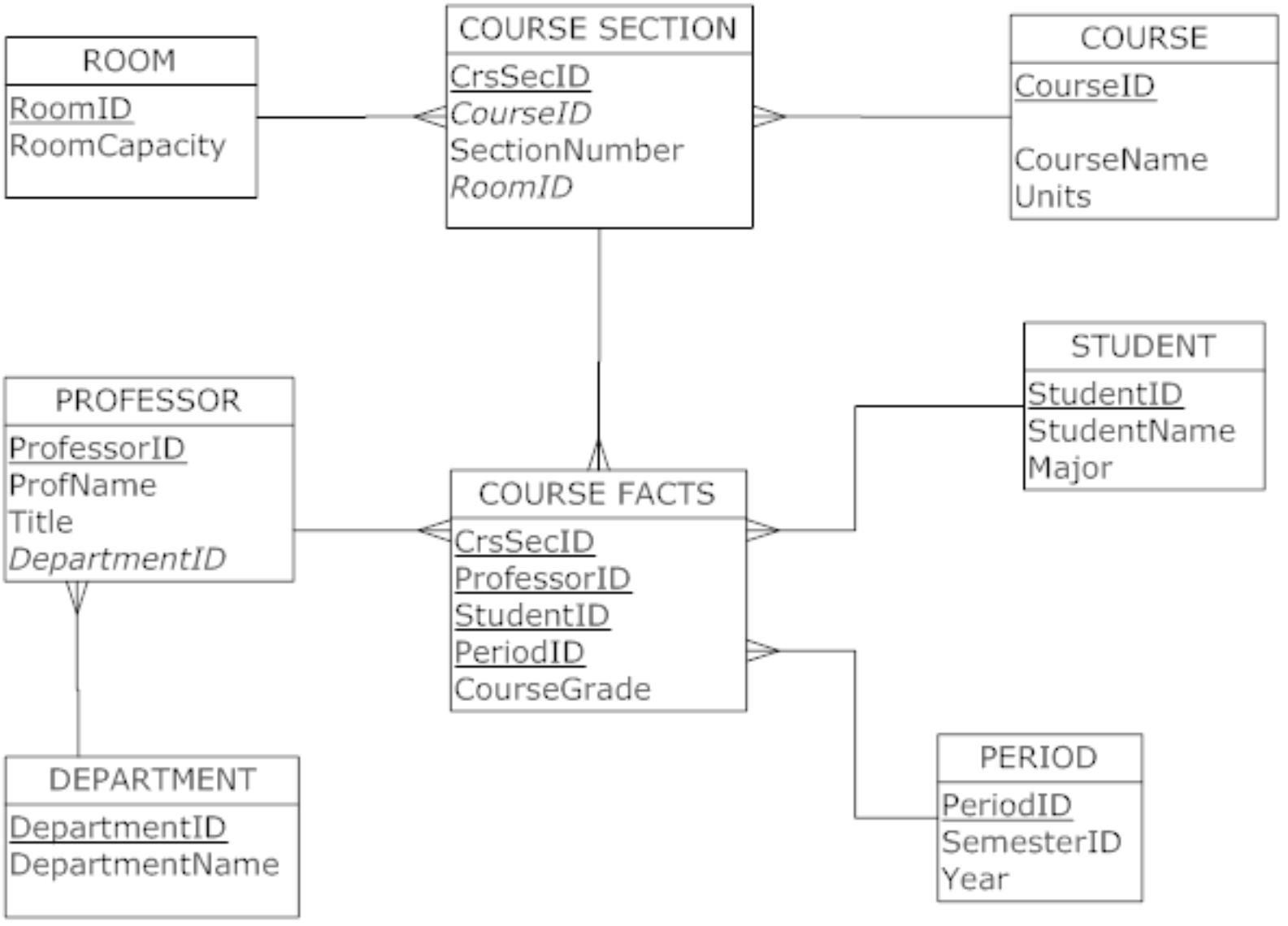
- There is a more complex version of the star schema called the snowflake schema.
- In the snowflake schema, each dimension table is normalized,
    - which may create additional tables attached to the dimension tables

Star Schema for A College Database

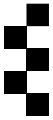


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Corresponding Snowflake Schema

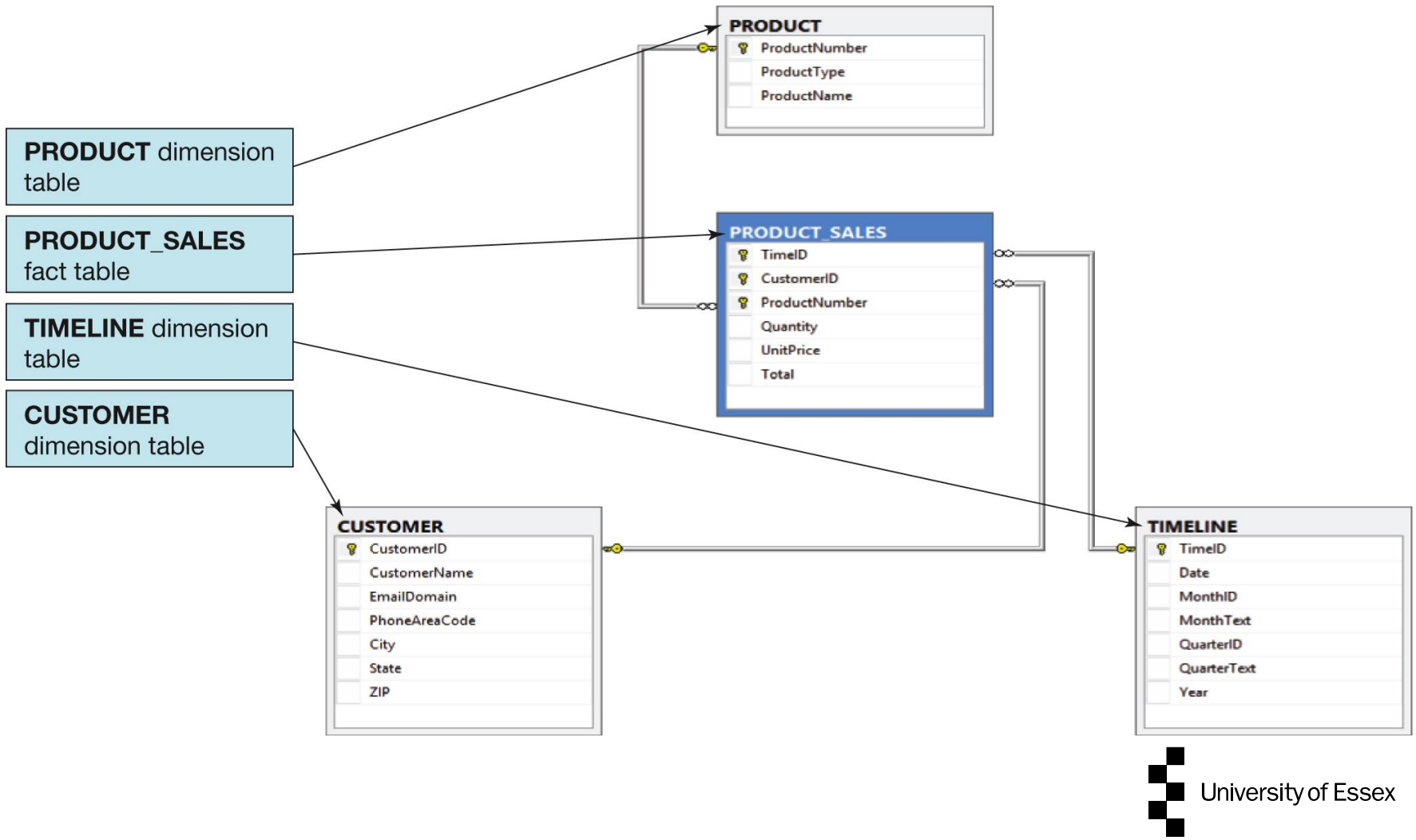


Implementation of a Data Warehouse



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The HSD_DW Star Schema (Figure 12-12)



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The HSD_DW SQL Create Table Statements (Figure 12-13)

```
CREATE TABLE TIMELINE(
      TimeID            Int              NOT NULL,
      Date              Date             NOT NULL,
      MonthID           Int              NOT NULL,
      MonthText         Char(15)         NOT NULL,
      QuarterID         Int              NOT NULL,
      QuarterText       Char(10)         NOT NULL,
      Year              Char(10)         NOT NULL,
      CONSTRAINT        TIMELINE_PK   PRIMARY KEY(TimeID)
      );

CREATE TABLE CUSTOMER(
      CustomerID        Int              NOT NULL,
      CustomerName      Char(75)         NOT NULL,
      EmailDomain       VarChar(100)     NOT NULL,
      PhoneAreaCode     Char(6)          NOT NULL,
      City              Char(35)         NULL,
      State             Char(2)          NULL,
      ZIP               Char(10)         NULL,
      CONSTRAINT        CUSTOMER_PK   PRIMARY KEY(CustomerID)
      );

CREATE TABLE PRODUCT(
      ProductNumber     Char(35)         NOT NULL,
      ProductType       Char(25)         NOT NULL,
      ProductName       VarChar(75)      NOT NULL,
      CONSTRAINT        PRODUCT_PK    PRIMARY KEY(ProductNumber)
      );

CREATE TABLE PRODUCT_SALES(
      TimeID            Int              NOT NULL,
      CustomerID        Int              NOT NULL,
      ProductNumber     Char(35)         NOT NULL,
      Quantity          Int              NOT NULL,
      UnitPrice         Numeric(9,2)     NOT NULL,
      Total             Numeric(9,2)     NULL,
      CONSTRAINT        SALES_PK      PRIMARY KEY
                              (TimeID, CustomerID, ProductNumber),
      CONSTRAINT        PS_TIMELINE_FK FOREIGN KEY(TimeID)
                              REFERENCES TIMELINE(TimeID)
                                 ON UPDATE NO ACTION
                                 ON DELETE NO ACTION,
      CONSTRAINT        PS_CUSTOMER_FK FOREIGN KEY(CustomerID)
                              REFERENCES CUSTOMER(CustomerID)
                                 ON UPDATE NO ACTION
                                 ON DELETE NO ACTION,
      CONSTRAINT        PS_PRODUCT_FK FOREIGN KEY(ProductNumber)
                              REFERENCES PRODUCT(ProductNumber)
                                 ON UPDATE NO ACTION
                                 ON DELETE NO ACTION
      );
```

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The HSD_DW Table Data (Figure 12-14)

	TimeID	Date	MonthID	MonthText	QuarterID	QuarterText	Year
1	43023	2017-10-15	10	October	3	Qtr3	2017
2	43033	2017-10-25	10	October	3	Qtr3	2017
3	43089	2017-12-20	12	December	3	Qtr3	2017
4	43184	2018-03-25	3	March	1	Qtr1	2018
5	43186	2018-03-27	3	March	1	Qtr1	2018
6	43190	2018-03-31	3	March	1	Qtr1	2018
7	43193	2018-04-03	4	April	2	Qtr2	2018
8	43198	2018-04-08	4	April	2	Qtr2	2018
9	43213	2018-04-23	4	April	2	Qtr2	2018
10	43227	2018-05-07	5	May	2	Qtr2	2018
11	43241	2018-05-21	5	May	2	Qtr2	2018
12	43256	2018-06-05	6	June	2	Qtr2	2018

(a) TIMELINE Dimension Table

	CustomerID	CustomerName	EmailDomain	PhoneAreaCode	City	State	ZIP
1	1	Jacobs, Nancy	somewhere.com	817	Fort Worth	TX	76110
2	2	Jacobs, Chantel	somewhere.com	817	Fort Worth	TX	76112
3	3	Able, Ralph	somewhere.com	210	San Antonio	TX	78214
4	4	Baker, Susan	elsewhere.com	210	San Antonio	TX	78216
5	5	Eagleton, Sam	elsewhere.com	210	San Antonio	TX	78218
6	6	Foxtrot, Kathy	somewhere.com	972	Dallas	TX	75220
7	7	George, Sally	somewhere.com	972	Dallas	TX	75223
8	8	Hullett, Shawn	elsewhere.com	972	Dallas	TX	75224
9	9	Pearson, Bobbi	elsewhere.com	512	Austin	TX	78710
10	10	Ranger, Terry	somewhere.com	512	Austin	TX	78712
11	11	Tyler, Jenny	somewhere.com	972	Dallas	TX	75225
12	12	Wayne, Joan	elsewhere.com	817	Fort Worth	TX	76115

(b) CUSTOMER Dimension Table

	ProductNumber	ProductType	ProductName
1	BK001	Book	Kitchen Remodeling Basics For Everyone
2	BK002	Book	Advanced Kitchen Remodeling For Everyone
3	BK003	Book	Kitchen Remodeling Dallas Style For Everyone
4	VB001	Video Companion	Kitchen Remodeling Basics
5	VB002	Video Companion	Advanced Kitchen Remodeling I
6	VB003	Video Companion	Kitchen Remodeling Dallas Style
7	VK001	Video	Kitchen Remodeling Basics
8	VK002	Video	Advanced Kitchen Remodeling
9	VK003	Video	Kitchen Remodeling Dallas Style
10	VK004	Video	Heather Sweeney Seminar Live in Dallas on 25-OCT-16

(c) PRODUCT Dimension Table

	TimeID	CustomerID	ProductNumber	Quantity	UnitPrice	Total
1	43023	3	VB001	1	7.99	7.99
2	43023	3	VK001	1	14.95	14.95
3	43033	4	BK001	1	24.95	24.95
4	43033	4	VB001	1	7.99	7.99
5	43033	4	VK001	1	14.95	14.95
6	43089	7	VK004	1	24.95	24.95
7	43184	4	BK002	1	24.95	24.95
8	43184	4	VK002	1	14.95	14.95
9	43184	4	VK004	1	24.95	24.95
10	43186	6	BK002	1	24.95	24.95
11	43186	6	VB003	1	9.99	9.99
12	43186	6	VK002	1	14.95	14.95
13	43186	6	VK003	1	19.95	19.95
14	43186	6	VK004	1	24.95	24.95
15	43186	7	BK001	1	24.95	24.95
16	43186	7	BK002	1	24.95	24.95
17	43186	7	VK003	1	19.95	19.95
18	43186	7	VK004	1	24.95	24.95
19	43190	9	BK001	1	24.95	24.95
20	43190	9	VB001	1	7.99	7.99
21	43190	9	VK001	1	14.95	14.95
22	43193	11	VB003	2	9.99	19.98
23	43193	11	VK003	2	19.95	39.90
24	43193	11	VK004	2	24.95	49.90
25	43198	1	BK001	1	24.95	24.95
26	43198	1	VB001	1	7.99	7.99
27	43198	1	VK001	1	14.95	14.95
28	43198	5	BK001	1	24.95	24.95
29	43198	5	VB001	1	7.99	7.99
30	43198	5	VK001	1	14.95	14.95
31	43213	3	BK001	1	24.95	24.95
32	43227	9	VB002	1	7.99	7.99
33	43227	9	VK002	1	14.95	14.95
34	43241	8	VB003	1	9.99	9.99
35	43241	8	VK003	1	19.95	19.95
36	43241	8	VK004	1	24.95	24.95
37	43256	3	BK002	1	24.95	24.95
38	43256	3	VB001	1	7.99	7.99
39	43256	3	VB002	2	7.99	15.98
40	43256	3	VK001	1	14.95	14.95
41	43256	3	VK002	2	14.95	29.90
42	43256	11	VB002	2	7.99	15.98
43	43256	11	VK002	2	14.95	29.90
44	43256	12	BK002	1	24.95	24.95
45	43256	12	VB003	1	9.99	9.99
46	43256	12	VK002	1	14.95	14.95
47	43256	12	VK003	1	19.95	19.95
48	43256	12	VK004	1	24.95	24.95

(d) PRODUCT_SALES Fact Table

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The HSD-DW SQL Query

```
/* *** SQL-Query-CH12-01 *** */
SELECT     C.CustomerID, C.CustomerName,
           P.ProductNumber, P.ProductName,
           SUM(PS.Quantity) AS TotalQuantity
FROM       CUSTOMER AS C, PRODUCT_SALES AS PS, PRODUCT AS P
WHERE      C.CustomerID = PS.CustomerID
  AND      P.ProductNumber = PS.ProductNumber
GROUP BY   C.CustomerID, C.CustomerName,
       P.ProductNumber, P.ProductName
ORDER BY   C.CustomerID, P.ProductNumber;
```

The HSD_DQ Query Results (Figure 12-15)

	CustomerID	CustomerName	ProductNumber	ProductName	TotalQuantity
1	1	Jacobs, Nancy	BK001	Kitchen Remodeling Basics For Everyone	1
2	1	Jacobs, Nancy	VB001	Kitchen Remodeling Basics	1
3	1	Jacobs, Nancy	VK001	Kitchen Remodeling Basics	1
4	3	Able, Ralph	BK001	Kitchen Remodeling Basics For Everyone	1
5	3	Able, Ralph	BK002	Advanced Kitchen Remodeling For Everyone	1
6	3	Able, Ralph	VB001	Kitchen Remodeling Basics	2
7	3	Able, Ralph	VB002	Advanced Kitchen Remodeling I	2
8	3	Able, Ralph	VK001	Kitchen Remodeling Basics	2
9	3	Able, Ralph	VK002	Advanced Kitchen Remodeling	2
10	4	Baker, Susan	BK001	Kitchen Remodeling Basics For Everyone	1
11	4	Baker, Susan	BK002	Advanced Kitchen Remodeling For Everyone	1
12	4	Baker, Susan	VB001	Kitchen Remodeling Basics	1
13	4	Baker, Susan	VK001	Kitchen Remodeling Basics	1
14	4	Baker, Susan	VK002	Advanced Kitchen Remodeling	1
15	4	Baker, Susan	VK004	Heather Sweeney Seminar Live in Dallas on 25-OCT...	1
16	5	Eagleton, Sam	BK001	Kitchen Remodeling Basics For Everyone	1
17	5	Eagleton, Sam	VB001	Kitchen Remodeling Basics	1
18	5	Eagleton, Sam	VK001	Kitchen Remodeling Basics	1
19	6	Foxtrot, Kathy	BK002	Advanced Kitchen Remodeling For Everyone	1
20	6	Foxtrot, Kathy	VB003	Kitchen Remodeling Dallas Style	1
21	6	Foxtrot, Kathy	VK002	Advanced Kitchen Remodeling	1
22	6	Foxtrot, Kathy	VK003	Kitchen Remodeling Dallas Style	1
23	6	Foxtrot, Kathy	VK004	Heather Sweeney Seminar Live in Dallas on 25-OCT...	1
24	7	George, Sally	BK001	Kitchen Remodeling Basics For Everyone	1
25	7	George, Sally	BK002	Advanced Kitchen Remodeling For Everyone	1
26	7	George, Sally	VK003	Kitchen Remodeling Dallas Style	1
27	7	George, Sally	VK004	Heather Sweeney Seminar Live in Dallas on 25-OCT...	2
28	8	Hullett, Shawn	VB003	Kitchen Remodeling Dallas Style	1
29	8	Hullett, Shawn	VK003	Kitchen Remodeling Dallas Style	1
30	8	Hullett, Shawn	VK004	Heather Sweeney Seminar Live in Dallas on 25-OCT...	1
31	9	Pearson, Bobbi	BK001	Kitchen Remodeling Basics For Everyone	1
32	9	Pearson, Bobbi	VB001	Kitchen Remodeling Basics	1
33	9	Pearson, Bobbi	VB002	Advanced Kitchen Remodeling I	1
34	9	Pearson, Bobbi	VK001	Kitchen Remodeling Basics	1
35	9	Pearson, Bobbi	VK002	Advanced Kitchen Remodeling	1
36	11	Tyler, Jenny	VB002	Advanced Kitchen Remodeling I	2
37	11	Tyler, Jenny	VB003	Kitchen Remodeling Dallas Style	2
38	11	Tyler, Jenny	VK002	Advanced Kitchen Remodeling	2
39	11	Tyler, Jenny	VK003	Kitchen Remodeling Dallas Style	2
40	11	Tyler, Jenny	VK004	Heather Sweeney Seminar Live in Dallas on 25-OCT...	2
41	12	Wayne, Joan	BK002	Advanced Kitchen Remodeling For Everyone	1
42	12	Wayne, Joan	VB003	Kitchen Remodeling Dallas Style	1
43	12	Wayne, Joan	VK002	Advanced Kitchen Remodeling	1
44	12	Wayne, Joan	VK003	Kitchen Remodeling Dallas Style	1
45	12	Wayne, Joan	VK004	Heather Sweeney Seminar Live in Dallas on 25-OCT...	1

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